

Adaptive Optimization Algorithms

Aleksandar Todorov

The Problem & Setup

Consider the stochastic optimization problem

$$\min_{x \in \mathbb{R}^d} F(x) = \mathbb{E}_{\xi}[F(x; \xi)]$$

with access only to stochastic gradients

$$g_n = \nabla F(x_n; \xi), \mathbb{E}[g_n \mid x_n] = \nabla F(x_n),$$

where F is possibly non-convex but L -smooth and
 $\|g_n\| \leq R$.

Recall the basic SGD update rule

$$x_{n+1} = x_n - \alpha_n g_n$$

with a $\mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$ convergence to a stationary point
under an appropriate step-size schedule.

Extended momentum variants (e.g. Polyak, Nestorov,
etc.) exist.

Implicit Assumption:

The learning rate is uniform for all optimization coordinates.

- SGD can be sensitive to gradient scaling and problem conditioning.
- Data problems have complex shapes.
- In function approximation from data, feature importance and frequency vary.

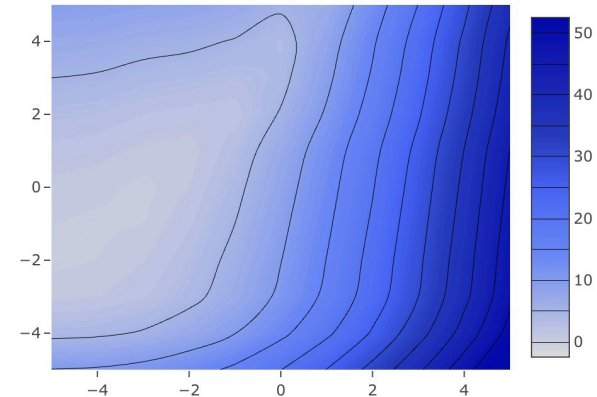


Figure 1: Optimizing along the horizontal axis is much easier than the vertical one.

What do we want?

- Larger steps for coordinates with smaller slopes.
- Smaller steps for coordinates with larger slopes.

Idea

Adjust the step size based on the history of each coordinate.

Let $g_n \in \mathbb{R}^d$ be the stochastic gradient at step n and write $g_{n,i}$ for coordinate i .

The coordinate-wise AdaGrad update is then:

$$x_{n+1,i} = x_{n,i} - \eta u_{n,i},$$

$$\text{where } v_{n,i} = v_{n-1,i} + g_{n,i}^2 \text{ and } u_{n,i} = \frac{g_{n,i}}{\sqrt{v_{n,i} + \varepsilon}}.$$

- The step size for each coordinate is adapted individually.
- Performs well in sparse-gradient settings.

AdaGrad Convergence (1)

AdaGrad converges to a critical point in the standard nonconvex sense:

$$\frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E} \|\nabla F(x_n)\|^2 \rightarrow 0$$

assuming a lower bounded smooth objective F and bounded stochastic gradients.

Descent-style lemma:

$$\mathbb{E}_{n-1} [\nabla_i F(x_{n-1}) u_{n,i}] \geq \frac{(\nabla_i F(x_{n-1}))^2}{2\sqrt{\varepsilon + \tilde{v}_{n,i}}} - 2R \mathbb{E}_{n-1} \left[\frac{g_{n,i}^2}{\varepsilon + v_{n,i}} \right],$$

where $\tilde{v}_{n,i} = v_{n-1,i} + \mathbb{E}_{n-1} [g_{n,i}^2]$ and $\|g_{n,i}\| \leq R$.

Together with smoothness, this gives

$$\begin{aligned} \mathbb{E}_{n-1}[F(x_n)] &\leq F(x_{n-1}) - \eta \sum_{i=1}^d \frac{(\nabla_i F(x_{n-1}))^2}{2\sqrt{\varepsilon + \tilde{v}_{n,i}}} + 2R\eta \sum_{i=1}^d \mathbb{E}_{n-1} \left[\frac{g_{n,i}^2}{\varepsilon + v_{n,i}} \right] + \frac{\eta^2 L}{2} \mathbb{E}_{n-1} \|u_n\|^2 \\ &\leq F(x_{n-1}) - \frac{\eta}{2R\sqrt{n}} \|\nabla F(x_{n-1})\|^2 + 2R\eta \sum_{i=1}^d \mathbb{E}_{n-1} \left[\frac{g_{n,i}^2}{\varepsilon + v_{n,i}} \right] + \frac{\eta^2 L}{2} \mathbb{E}_{n-1} \|u_n\|^2. \end{aligned}$$

AdaGrad Convergence (2)

Taking a full expectation, summing, telescoping, and using lower-boundedness, we obtain

$$\frac{\eta}{2R} \sum_{n=1}^N \frac{1}{\sqrt{n}} \mathbb{E} \|\nabla F(x_{n-1})\|^2 \leq F(x_0) - F_* + 2R\eta \sum_{n=1}^N \sum_{i=1}^d \mathbb{E} \left[\frac{g_{n,i}^2}{\varepsilon + v_{n,i}} \right] + \frac{\eta^2 L}{2} \sum_{n=1}^N \mathbb{E} \|u_n\|^2.$$

Technical Lemma

For a nonnegative sequence a_n and $b_n = \sum_{k=1}^n a_k$, we have $\sum_{n=1}^N \frac{a_n}{\varepsilon + b_n} \leq \ln\left(1 + \frac{b_N}{\varepsilon}\right)$.

Applying this on the last two terms and rearranging gives

$$\frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E} \|\nabla F(x_n)\|^2 \leq \frac{2R(F(x_0) - F_*)}{\eta\sqrt{N}} + \frac{1}{\sqrt{N}} (4dR^2 + \eta dRL) \ln\left(1 + \frac{NR^2}{\varepsilon}\right),$$

which gives a $\mathcal{O}\left(\frac{d \log N}{\sqrt{N}}\right)$ convergence to a stationary point.

Can We Do Better?

v_n grows monotonically, decay can be too aggressive.

RMSProp: use an exponentially decaying moving average

$$v_n = \beta v_{n-1} + (1 - \beta) g_n^2,$$

where $0 < \beta < 1$ is a decay rate parameter.

Intuition

- The denominator $\sqrt{v_n^2 + \varepsilon}$ adapts to the recent magnitude of gradients for each parameter.
- If a parameter's gradients become small after initially being large, v_n^2 will decrease.

Convergence

- The asymptotic behavior depends strongly on β .
- Can be made $\mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$ under diminishing step sizes and when $\beta \rightarrow 1$, behaves like AdaGrad.

Adam (Adaptive Moment Estimation) = RMSProp + Momentum + Bias Correction

Adam maintains:

- First momentum vector (m): an exponentially decaying average of past gradients

$$m_n = \beta_1 m_{n-1} + (1 - \beta_1) g_n.$$

- Second moment vector (v): an exponentially decaying average of past squared gradients

$$v_n = \beta_2 v_{n-1} + (1 - \beta_2) g_n^2.$$

Update bias:

- During early optimization, m_n and v_n are biased towards zero.

- The bias corrected estimates become

$$\hat{m}_n = \frac{m_n}{1 - \beta_1^n},$$

$$\hat{v}_n = \frac{v_n}{1 - \beta_2^n}.$$

- $(1 - \beta_1^n)$ and $(1 - \beta_2^n)$ approach 1 as n increases.

The update rule then becomes

$$x_{n+1} = x_n - \eta \frac{\hat{m}_n}{\sqrt{\hat{v}_n} + \varepsilon}.$$

Adam Benefits & Downsides

- Combines benefits (adaptive moments with momentum).
- Robust, good default settings, less tuning needed.
- Performs extremely well empirically.
- No convergence guarantee, even in the convex setting.
- Can construct ill-posed examples for Adam.
- Further modifications and theoretical fixes but less used (AMSGrad, Adamax, Nadam).

Time For a Short Live Demo!

SGD vs. AdaGrad vs. RMSProp vs. Adam

References

- Ward, R., Wu, X., & Bottou, L. (2019). AdaGrad Stepsizes: Sharp Convergence Over Nonconvex Landscapes. In K. Chaudhuri & R. Salakhutdinov (Eds.), Proceedings of the 36th International Conference on Machine Learning: Vol. 97. Proceedings of the 36th International Conference on Machine Learning. <https://proceedings.mlr.press/v97/ward19a.html>
- Défossez, A., Bottou, L., Bach, F., & Usunier, N. (2022,). A Simple Convergence Proof of Adam and Adagrad. <https://arxiv.org/abs/2003.02395>
- Duchi, J., Hazan, E., & Singer, Y. (2011). Adaptive Subgradient Methods for Online Learning and Stochastic Optimization. J. Mach. Learn. Res., 12(null), 2121–2159.
- ApX Machine Learning. (2026,). Model Regularization and Optimization in Deep Learning. ApX Machine Learning. <https://apxml.com/courses/deep-learning-regularization-optimization>
- Kingma, D. P., & Ba, J. (2017,). Adam: A Method for Stochastic Optimization. <https://arxiv.org/abs/1412.6980>
- Reddi, S. J., Kale, S., & Kumar, S. (2019,). On the Convergence of Adam and Beyond. <https://arxiv.org/abs/1904.09237>